INJECTIVITY OF THE RESTRICTION OF DERIVATIONS FORCES SURJECTIVITY OF THE RING MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-defos.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the translation of nonsurjectivity into the nonvanishing of the relative cotangent quotient, the explicit verification of the Leibniz rule for the quotient map, and the deduction of a nonzero derivation trivial on the source ring.

Editorial note. The label lemma-derivations-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source formal-defos.tex –label lemma-derivations-surjective –start 1219 –end 1248 –local-id 28025 returned RESULT=OK (L28025 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-formal-defos/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Detecting surjectivity on the relative cotangent space. Same chapter, the lemma that a map in the completed category is surjective exactly when its relative cotangent space vanishes. Supplies the nonvanishing of the relative cotangent space when f is not surjective, hence the nonvanishing of the quotient Q.
- Structure of the quotient by the subring and the square of the maximal ideal. Standard properties of the quotient of a local algebra by the image of a ring map plus the square of its maximal ideal, making it a vector space over the residue field. Supplies the identification of Q as a k-vector space and as an S-module through the residue field.

Source reference. Original reference 06GZ: sources/stacks-formal-defos/pinned-original-source.tex, source lines 1086–1086. Prerequisite pointer invoked at proof line(s) 1229; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : R \to S$ be a morphism of $\widehat{\mathcal{C}}_\Lambda$. If $Der_\Lambda(S, k) \to Der_\Lambda(R, k)$ is injective, then f is surjective.*

Date: stacks-project:formal-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. If f is not surjective, then $\mathfrak{m}_S/(\mathfrak{m}_R S + \mathfrak{m}_S^2)$ is nonzero by Lemma 06GZ. Then also $Q = S/(f(R) + \mathfrak{m}_R S + \mathfrak{m}_S^2)$ is nonzero. Note that Q is a $k = R/\mathfrak{m}_R$-vector space via f. We turn Q into an S-module via $S \to k$. The quotient map $D : S \to Q$ is an R-derivation: if $a_1, a_2 \in S$, we can write $a_1 = f(b_1) + a_1'$ and $a_2 = f(b_2) + a_2'$ for some $b_1, b_2 \in R$ and $a_1', a_2' \in \mathfrak{m}_S$. Then b_i and a_i have the same image in k for $i = 1, 2$ and

$$\begin{aligned} a_1 a_2 &= (f(b_1) + a_1')(f(b_2) + a_2') \\ &= f(b_1)a_2' + f(b_2)a_1' \\ &= f(b_1)(f(b_2) + a_2') + f(b_2)(f(b_1) + a_1') \\ &= f(b_1)a_2 + f(b_2)a_1 \end{aligned}$$

in Q which proves the Leibniz rule. Hence $D : S \to Q$ is a Λ-derivation which is zero on composing with $R \to S$. Since $Q \neq 0$ there also exist derivations $D : S \to k$ which are zero on composing with $R \to S$, i.e., $\mathrm{Der}_\Lambda(S, k) \to \mathrm{Der}_\Lambda(R, k)$ is not injective. $\square$